ChemRisk-AI: Mechanism-Aware Mitigation Decision Support

A weak-supervision framework for chemical incompatibility screening, corrected v1.2b RTD recommendation, and direction-aware tank/service-level review

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Reader Orientation: What This Project Is About

This project sits at the intersection of chemical process engineering, process safety, and applied machine learning. It does not try to replace chemical experts or perform first-principles reaction modeling. Instead, it builds a decision-support system that helps engineers decide which safeguards are appropriate when two incompatible chemicals could be accidentally mixed.

The key engineering decision in the Phase 1 MVP is whether a temperature sensor, called an RTD, is a useful and creditable secondary safeguard for a given chemical mixing scenario and, ultimately, for the storage tank that holds one of those chemicals. The design now explicitly separates pair-level incompatibility from direction-specific tank scenarios, because A entering Tank B can differ from B entering Tank A when inventory, concentration, and dilution change the credible mechanism.

Term / Concept	Definition
Chemical incompatibility screening	A structured review of what could happen if two chemicals are accidentally mixed, such as heat generation, gas evolution, toxic vapor release, fire, pressure rise, or corrosion.
C1-C6 severity scale	A consequence ranking scale used in the screening workbook. C1-C4 represent lower to moderate consequences; C5-C6 represent high-severity outcomes that typically require stronger safeguards or review.
C5/C6	The high-severity band. In this project, C5 and C6 are often treated similarly from a mitigation trigger standpoint, so a C6-to-C5 change may not materially change the required safeguard.
RTD	Resistance Temperature Detector. A sensor installed in a tank to measure temperature. It is useful only when the hazardous event produces a detectable bulk temperature rise in time for action.
CAMEO-style hazard flags	Conservative theoretical indicators such as heat generation, gas generation, toxic gas, fire, or explosive potential due to the reaction of two chemicals. These flags are useful for screening but can overstate what is physically credible in a real process setting.
Gold-set audit	A manually reviewed subset of chemical pairs used to check whether the weak labeling rules agree with expert engineering judgment.
Weak supervision / labeling functions	A method for creating training labels from multiple partial rules. Each rule votes only when its conditions apply; otherwise it abstains. The votes are aggregated into a weak label for model training.
Pair-level vs. tank-level decision	The dataset evaluates chemical pairs, but the physical RTD installation decision is made for a storage tank that normally contains one chemical. The system therefore aggregates pair-level predictions into a tank-level recommendation.
Directional scenario	A tank-context scenario that defines which chemical is stored as the majority inventory and which incompatible chemical enters as the abnormal contaminant, such as A_into_B or B_into_A.
Inventory context	Tank level, stored quantity, concentration, dilution, and order-of-addition information that can affect heat release, gas evolution, kinetics, and whether an RTD detects a useful bulk temperature signal.
Review routing	A conservative decision state used when the model or labeling functions do not have enough evidence to force a binary RTD recommendation.

1. Business Problem & Context

Chemical manufacturing sites intentionally use conservative screening methods when performing process and project hazard analyses because the cost of missing a severe incompatibility can be high. However, conservative screening can also create over-scoped mitigation designs, repeated lab testing, and decision fatigue when every high-severity flag is treated as if it requires the same hardware solution.

The original project hypothesis was that conservative screening would frequently overstate severity. The completed gold-set audit showed a more nuanced result: most severity ratings remained unchanged or moved within the same high-severity band, such as C6 to C5. Because C5 and C6 both generally trigger mitigation review, those changes do not always reduce scope.

The stronger and more actionable problem is mitigation selection. A scenario may truly be high severity, but an RTD may still be the wrong safeguard if the dominant hazard is toxic gas, pressure generation, fire, very fast decomposition, or a gel/sludge/fouling mechanism that prevents reliable temperature detection.

A related asset-level issue is directionality: the same pair can have different practical consequences depending on which material is stored in the tank, which material is introduced, and the tank inventory at the time of the abnormal transfer. The MVP therefore treats directionality as a tank-scenario and review-routing layer rather than hiding it inside the pair-level classifier.

Original Framing	Revised Framing	Why the Revision Matters
Can ML reduce conservative severity ratings?	Can ML identify when RTD installation is actually useful, creditable as a safeguard, and practical?	This aligns the model target with the observed data signal and with a real engineering decision that affects capital scope.

2. System Architecture

The system follows a piped-inference architecture. In plain language, this means the system transforms structured chemical-pair data into intermediate mechanism signals, aggregates weak supervision votes, trains a tabular model, and then converts pair-level predictions into direction-aware, tank-level recommendations.

The Phase 1 MVP is intentionally tabular and interpretable. It uses the public structured dataset, gold-set audits, calibrated weak-labeling functions, leakage-safe baseline models, and a directionality-aware tank-scenario layer to build a credible proof of concept. Phase 2, which would use SDS/NLP extraction and richer site data, remains future scope.

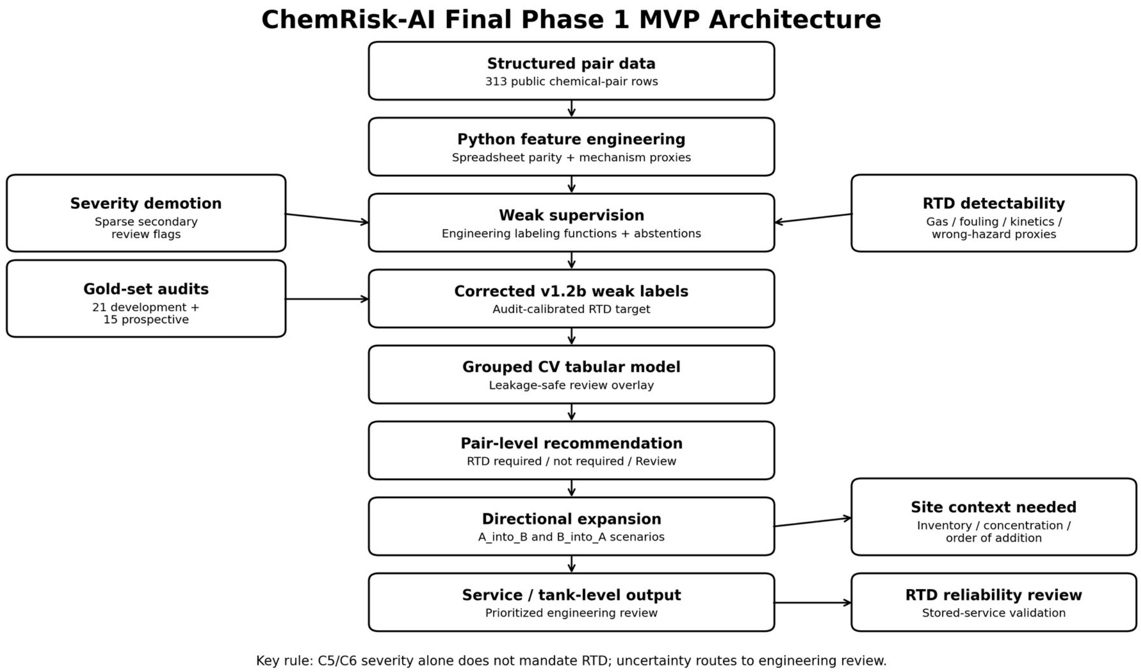

Figure 1. Final ChemRisk-AI Phase 1 MVP architecture. The final workflow uses mechanism-aware weak labels, audit-calibrated v1.2b RTD decisions, leakage-safe grouped cross-validation, model-assisted review routing, directional A_into_B / B_into_A scenario expansion, and stored-service RTD reliability review before final service-level recommendations.

2.1 Phase 1 Objective - Capstone MVP

Develop a reliable, interpretable, tabular decision-support model that:

- Predicts pair-level RTD usefulness / RTD required vs. not required.
- Explains the dominant control failure mode when RTD is not appropriate.
- Provides severity context and rule-based flags for the small subset of decision-relevant severity demotion candidates below C5.
- Aggregates pair-level predictions to direction-aware tank-level RTD recommendations using stored chemical service reliability review and A_into_B / B_into_A scenario logic.
- Flags directionality-sensitive or inventory-dependent cases for engineering review when tank level, concentration, or order-of-addition could change the credible mechanism.
- Quantifies confidence and routes uncertain cases to peer review or lab validation.

2.2 Phase 2 Future Extension

Phase 2 would expand the system beyond known structured rows by extracting chemical features from Safety Data Sheets (SDSs). This could include incompatible materials, decomposition products, pH, and physical-property clues. This extension is useful for future generalization, but it is not required for the Phase 1 capstone MVP.

2.3 Current MVP vs. Future Product

MVP means Minimum Viable Product. In this project, the Phase 1 MVP is the smallest working version that demonstrates the core decision-support workflow: structured chemical-pair screening, mechanism-aware weak supervision, audited calibration, leakage-safe modeling, model-assisted review routing, directionality expansion, and service/tank-level recommendation.

The current MVP is useful as a reproducible workflow architecture and portfolio demonstration. It is not presented as a broadly deployable chemical-classification product that can accept arbitrary SDS files or arbitrary site inventories without additional validation.

The future product would add SDS ingestion, chemical identity normalization, confidence-scored generic class mapping, tank inventory mapping, and multi-site validation. This distinction keeps the Phase 1 claims realistic while showing a clear path toward industrial deployment.

- Phase 1 - completed: weak-supervised MVP using structured chemical-pair data, audited calibration, model-assisted review, and direction-aware service/tank aggregation.
- Phase 2 - future: synthetic demo app where users can select generic chemical classes and view recommendation explanations.
- Phase 3 - future: SDS ingestion and chemical class mapping using NLP/LLM-assisted extraction plus deterministic validation checks.
- Phase 4 - future: multi-site validation with larger audited datasets, tank maps, inventory context, and outcome/lab data where available.

2.4 Why Not Just Use a General-Purpose LLM?

A general-purpose LLM could be valuable in a future ChemRisk-AI architecture, especially for extracting structured information from Safety Data Sheets and mapping raw text into chemical identity, pH, incompatibility, oxidizer/reducer, decomposition, flammability, and physical-property clues.

However, the final safeguard recommendation should not rely on a general LLM alone. Safety-critical recommendations require a controlled schema, reproducible decision logic, audit traceability, versioned calibration, leakage-safe evaluation, explicit uncertainty handling, and human-in-the-loop engineering review.

The safest product architecture would combine both approaches: an LLM or SDS parser for unstructured information extraction, followed by ChemRisk-AI style mechanism-aware rules, calibrated tabular modeling, direction-specific scenario expansion, and engineering review routing.

3. Data Readiness & Current Dataset Reality

The current public dataset is sufficient for a weak-supervised, tabular MVP focused on RTD recommendation and mitigation selection. It is not sufficient for a full mechanistic chemistry predictor, a robust six-class severity model, or a complete kinetics model that resolves direction, concentration, and tank inventory without site context. This distinction is important because it keeps the project credible.

Item	Current Status / Interpretation
Rows available	Approximately 313 public chemical-pair rows. The final notebook expands these into 626 directional scenarios using A_into_B and B_into_A views.
Gold-set status	36 audited rows were used for MVP calibration and error analysis: 21 initial development audits plus 15 prospective stress-test audits selected after v1.1 was frozen. The full 50-row target remains useful for stronger validation.
Strongest signal	RTD usefulness / mitigation misapplication, especially whether a high-severity scenario produces a detectable bulk-temperature signal that makes RTD creditable.

Weakest signal	Full six-class severity prediction, broad severity-demotion model training, and first-principles kinetics without lab or site data.
Main limitation	No direct measured kinetics, heat of reaction, viscosity, phase behavior, tank level, concentration, order-of-addition, transfer rate, or incident-outcome data.
Current implementation status	Final Colab implementation is complete for Phase 1 MVP: Python parity checks, corrected v1.2b weak-label layer, 271 leakage-safe training rows, grouped CV, model-assisted review overlay, directional scenario expansion, service-level aggregation, enhanced retained-RTD chemistry-basis explanations, model card, summary tables, and GitHub figures.
Directionality status	Directionality is handled as a separate scenario-expansion and review-routing layer. Pair-level output is not treated as a final tank decision until stored/incoming direction and stored-service reliability are reviewed.
Validation status	The 36-row audit set supports MVP calibration and active-learning stress testing, not statistically powered validation. Final model results should be presented as calibrated decision-support evidence, not production qualification.

3.1 What the 21-Row Gold-Set Audit Established

- Six of twenty-one severity ratings changed in the audit.
- Four of those six changed from C6 to C5, which usually does not change the mitigation trigger because both remain in the high-severity band.
- Two changed from C6 to C2, which is decision-relevant. Those cases were associated with thermal buffering/dilution and nonhazardous exotherm behavior.
- The dominant value opportunity is not broad severity demotion. It is identifying whether RTD installation is useful, ineffective, or the wrong safeguard type.

3.2 Recent Severity-Demotion Finding

The severity-demotion signal is real but sparse. In the public workbook, the active severity-demotion flags currently produce only a small number of candidate rows. That is useful for triage and engineering review, but not enough to justify a separate supervised severity-demotion model in the Phase 1 MVP.

3.3 Frozen v1.1, Prospective Audit Cycle, and Corrected v1.2b

After spreadsheet-to-Python parity was achieved, targeted gold-set diagnostics exposed a small number of RTD false positives and one false negative. These were first addressed in a calibrated v1.1 weak-label layer without overwriting the spreadsheet-parity baseline. The prospective 15-row audit then exposed additional gas/pressure, fast-kinetics, and phase/fouling failure modes that justified a corrected v1.2b layer.

The 15 prospective audit rows were selected after freezing v1.1 using active-learning criteria: model disagreement, out-of-fold uncertainty, moderate-confidence single-LF votes, minority-class RTD-required cases, and mechanism coverage gaps. Once those rows were audited and used to update v1.2b, the combined 36-row set became a development/calibration set rather than independent validation.

3.4 Final Phase 1 Dataset and Validation Status

- The final Phase 1 implementation uses 313 public chemical-pair rows and expands them into 626 direction-specific scenarios for tank/service-level review.
- The combined audit set contains 36 reviewed rows: 31 audited RTD-not-required cases and 5 audited RTD-required cases. This is enough for MVP calibration, but not enough for statistically powered validation.

- The corrected v1.2b weak-label layer matched the combined audited development/calibration set with 30 true negatives, 1 false positive, 0 false negatives, and 5 true positives. This supports the mechanism logic but should not be presented as production validation.

4. Weak Supervision Framework

Because high-consequence incident data is sparse, the project uses weak supervision instead of traditional supervised learning with clean ground-truth labels. Labeling functions are engineering-informed rules that vote only when their conditions apply. If a rule does not have enough evidence, it abstains rather than forcing a vote.

Abstentions are ignored during aggregation. For example, if three active rules vote [1, 0, 1] for RTD installation and five other rules abstain, the unweighted probability of RTD required is 2/3. Abstentions are not averaged as negative values.

4.1 Secondary Severity-Demotion Review Flags

Severity calibration is retained only as secondary context. It is not the main model target. The current workbook treats severity demotion as a sparse rule-based review flag rather than a supervised learning target.

Severity LF	Current v1 Status	Use in Project
SLF1_ThermalBuffer_Demotion	Candidate review flag. Broad proxy: high severity plus diluent / thermal-buffer indication. Current public workbook restores this as a candidate flag, not a validated model target.	Flags potential cases where dilution or thermal mass may make the initial severity overly conservative.
SLF2_NonHazardousExotherm_Demotion	High-value candidate review flag. Current proxy identifies sludge/gel/polymer/precipitation-style nonhazardous-exotherm patterns. Low sample size but validated on the limited audited rows where it voted.	Flags cases where heat may be present but not hazardous or runaway-capable.
SLF3_OpenAirExposure_Demotion	Retired from active v1 aggregation due to poor early gold-set precision.	Mentioned only as future work; not used in the current workbook or Colab target.

These severity-demotion flags should be reported in Colab as candidate review opportunities. They should not be used to train a separate severity-demotion classifier unless more positive examples are audited later.

4.2 Primary Mitigation-Decision Layer

The primary layer predicts whether an RTD is required and creditable as a thermal safeguard. The active labeling function (LF) system intentionally separates the broad baseline rule from the final vote. The broad rule votes for RTD installation for C5/C6 reactions that generate heat, but the rule is retained only as a comparator because the gold-set audit showed it over-recommends RTDs.

Labeling Function	Definition
LF1_NoHeat_LowSeverity_NoRTD	Votes no RTD when the row is below the high-severity band or has no heat-generation flag.
LF2_BulkLiquidNeutralization_RTD	Votes RTD required for clean, high-severity bulk-liquid acid/base exotherms with no obvious fouling, gas, or gel proxy.
LF3_ToxicGasDominant_NoRTD	Votes no RTD when the likely dominant hazard is toxic gas or vapor release, not useful

	bulk temperature rise.
LF4_DetectabilityFailure_NoRTD	Votes no RTD when classes imply gel, sludge, precipitation, phase separation, gas shrouding, or fouling.
LF5_FastPressureDecomp_NoRTD	Votes no RTD when fast gas/pressure/decomposition mechanisms are likely to outrun RTD-based prevention.
LF6_OxidizerOrganic_NoRTD_Candidate	Candidate no-RTD vote for oxidizer-organic/fire/decomposition patterns; requires continued gold-set validation.
LF7_Baseline_SevereHeat_RTD	Baseline comparator only. It is excluded from final weak voting because it is too broad.
LF9_Thermal_Default_RTD	Controlled positive RTD pathway: severe + heat + no active negative failure-mode proxy. In v1.2b, LF9 is recomputed after calibrated negative rules and protected positive RTD patterns are applied.
LF8_Diluent_NoRTD_Candidate	Candidate no-RTD vote for severe heat rows where dilution / thermal buffering and absence of toxic gas or explosive potential suggest the exotherm may not require RTD as the primary safeguard.
LF10_OxidizingAcidChelant_NoRTD	Calibrated v1.1 no-RTD vote for oxidizing acid + chelant scenarios where gas, pressure, or wrong-hazard-type behavior dominates over useful bulk temperature detection.
LF11_AcidicSurfactantGlycolEther_NoRTD	Calibrated v1.1 no-RTD vote for acidic anionic surfactant + glycol ether solvent cases where detectability or wrong-safeguard concerns are more credible than an RTD-creditable bulk thermal event.
LF12_AcidicSurfactantVolatileBase_RTD	Calibrated v1.1 positive RTD vote for severe heat-generating acidic anionic surfactant + volatile/ammoniacal base neutralization where the audited mechanism supports an RTD-creditable thermal safeguard.
LF13_NonthermalDominance_NoRTD	Corrected v1.2b no-RTD vote for high-severity cases where gas, pressure, toxic-vapor, fire, or wrong-hazard-type behavior dominates over useful bulk temperature detection.
LF14_PhaseBehaviorFouling_NoRTD	Corrected v1.2b no-RTD vote for phase separation, gel, sludge, polymer/emulsion, silicate, surfactant, metal-complex, or fouling patterns that may prevent reliable RTD credit.
LF15_FastPressureKinetics_NoRTD	Corrected v1.2b no-RTD vote for oxidizer/decomposition, hypochlorite/acid, amine/oxidizer, or fast gas/pressure mechanisms that may outrun RTD-based prevention.
LF16_ProtectedBulkAcidBase_RTD	Corrected v1.2b positive RTD vote that protects audited bulk-liquid acid/base exotherm patterns from being over-suppressed by broad no-RTD proxies.

4.3 Probabilistic Aggregation

- Collect all nonblank votes from the active LF set.
- Ignore abstentions.
- Compute weak_vote_prob as the mean or weighted mean of active votes.
- Assign RTD required, RTD not required, or Review based on probability and conflict rules.
- Measure LF performance against the audited gold set using coverage, precision when voting, and conflict rate.

4.4 Calibrated v1.1 and Corrected v1.2b Weak-Label Layers

The Colab notebook preserves the original spreadsheet-parity LF reconstruction using py_lf... columns, then creates calibrated layers using cal_lf... and v12b_lf... columns. This versioning is intentional: parity proves reproducibility, v1.1 documents targeted corrections from the first development audit, and v1.2b documents the prospective-audit corrections without silently changing earlier layers.

The corrected v1.2b layer is the final Phase 1 MVP weak-label layer. It reduces over-recommendation for gas/pressure, fast-kinetics, and phase/fouling mechanisms while preserving known RTD-creditable bulk-liquid acid/base exotherm cases. The model is then used as a review overlay, not as an automatic override of mechanism-aware rules.

5. Mitigation Feasibility / RTD Detectability

An RTD is not automatically useful just because a reaction generates heat. It is useful only when the event produces a bulk temperature signal that is detectable in time for action and the stored material does not make the instrument unreliable.

The Phase 1 dataset does not contain direct measured viscosity, mixing quality, heat-transfer coefficient, thermowell fouling, bulk-vs-localized temperature response, tank level, concentration, or direction/order-of-addition data for every chemical pair. Therefore, the system uses engineered proxy indicators derived from chemical class, reaction type, hazard flags, and a separate tank-scenario review layer.

The Phase 1 RTD detectability proxies include:

• Bulk-liquid exotherm proxy:

inferred from reaction type, heat-generation flag, high-severity context, and chemical-class combinations consistent with clear-liquid acid/base exotherms.

• Detectability-failure proxy:

inferred from chemical-class combinations associated with gel formation, sludge, precipitation, phase separation, polymer/emulsion instability, or thermowell fouling.

• Gas / pressure / sensor-lag proxy:

inferred from gas-generation, explosive-potential, oxidizer/decomposition chemistry, and sparse reaction-speed information where available.

• Wrong-hazard-type proxy:

inferred when toxic gas, fire, pressure, corrosion, or decomposition appears to dominate over bulk thermal runaway.

• Stored-service RTD reliability review:

determined using a separate chemical-service lookup that flags stored services where coating, fouling, foaming, phase separation, poor mixing, or similar reliability concerns should be reviewed before crediting RTD.

• Directionality / inventory sensitivity proxy:

inferred when acid/base neutralization, oxidizer/reducer chemistry, hypochlorite/acid or amine chemistry, silicate gel formation, polymer/emulsion destabilization, metal complexation, gas generation, or unknown reaction speed could make A_into_B different from B_into_A.

• Valid RTD-use proxy:

inferred when a credible severe heat-generating scenario is present and no toxic-gas, detectability-failure, fast-pressure, or stored-service reliability concern proxy dominates.

5.1 RTD Recommendation Logic

RTD is recommended only if all of the following are true:

- A credible exothermic reaction is present.
- The exotherm is hazardous, not merely heat-generating.
- The event is expected to produce a bulk temperature rise detectable by RTD.
- No dominant failure mode makes RTD ineffective, such as toxic-gas dominance, gas shrouding, gel/sludge/fouling, phase separation, or reaction kinetics too fast for preventive action.
- At the tank level, the stored chemical service has no unresolved RTD reliability concern for the relevant direction of the abnormal transfer.
- Directionality-sensitive or inventory-dependent cases are routed to engineering review unless sufficient site context resolves the stored/incoming-material scenario.

If any of these conditions fail, RTD is not recommended as the primary thermal safeguard. Other safeguards may still be required.

6. Tank-Level RTD Aggregation Layer

The source data is chemical-pair based, but RTD installation is an asset-level decision. A tank normally stores one chemical service, while the incompatible chemical is introduced by an abnormal transfer or unloading scenario. Therefore, pair-level model output must be expanded into direction-aware tank scenarios before final recommendations are made.

The tank-level decision combines three inputs:

- Pair-level weak/model prediction: whether the pair has a credible RTD-creditable thermal mechanism at the screening level.
- Stored-service RTD reliability review lookup: whether the chemical normally stored in the tank is compatible with reliable temperature measurement.
- Aggregation across all relevant incompatible pairs and directions for that stored chemical, including A_into_B and B_into_A scenario expansion where appropriate.

For example, a polymer, surfactant, silicate, or wax-emulsion tank may require RTD reliability review even if one pair-level scenario includes heat, because normal service coating, fouling, phase behavior, gas entrainment, or mixing limitations could make a thermowell measurement less creditable. Similarly, an acid/base or oxidizer/reducer pair may need directionality review because the mechanism can depend on which material is present as the majority tank inventory. In those cases, the system recommends engineering review rather than crediting RTD as the primary safeguard without site context.

6.1 Directionality and Inventory Context

Directionality is handled as a scenario layer instead of being embedded as an unsupported assumption in the pair-level classifier. For each incompatible pair, the tank-level layer can create two conceptual scenarios: A_into_B, where chemical B is the stored/major inventory and A is the incoming abnormal contaminant; and B_into_A, where chemical A is stored and B enters abnormally.

This matters because majority/minority material, tank level, concentration, dilution, heat capacity, local mixing, and pH endpoint can change the credible mechanism. A reaction that is RTD-creditable when a small amount of acid enters a large base tank may not behave the same way when the order is reversed, especially for gas evolution, fast decomposition, precipitation, gel formation, or localized exotherms.

In Phase 1, the public dataset does not contain enough site-specific information to calculate these effects. The model therefore uses conservative directionality-sensitive flags and review routing. If either direction appears RTD-creditable, the case remains a candidate RTD scenario. If either direction is materially direction-sensitive and unresolved, the final tank recommendation should include engineering review rather than a fully automated binary conclusion.

7. Final Colab Implementation Workflow

The final Colab notebook implements the Phase 1 workflow end to end. The public GitHub version includes the notebook structure, model card, summary tables, figures, and documentation, while private audit workbooks are intentionally omitted from the repository.

1. Load the public structured chemical-pair dataset and private audit workbooks for development analysis.

2. Validate schema, data types, missing values, and high-level hazard distributions.

3. Rebuild helper features and mechanism-aware labeling functions in Python rather than relying on Excel recalculation.

4. Verify spreadsheet-to-Python parity for helper features and RTD labeling functions.

5. Evaluate LF coverage, precision when voting, conflict behavior, and initial audit disagreement.

6. Freeze calibrated v1.1, select 15 prospective audit rows using model disagreement and uncertainty, then evaluate before creating v1.2b.

7. Create corrected v1.2b weak labels that reduce gas/pressure, fast-kinetics, and phase/fouling over-recommendation while protecting RTD-creditable bulk acid/base cases.

8. Train leakage-safe grouped cross-validation baseline models using canonical chemical-pair groups, class weighting, and evidence-tier sample weights.

9. Use the primary evidence-weighted logistic model as a review overlay: disagreements route to Review rather than silently overriding engineering rules.

10. Expand pair-level recommendations into A_into_B and B_into_A directional scenarios, then aggregate by stored chemical service as a Phase 1 proxy for tank-level recommendation.

7.1 Row-Level Explanation Layer

The final public notebook now includes a row-level explanation layer. This layer is designed to answer why a pair-level or directional recommendation was made, rather than leaving the output as a model score or unexplained label.

For each pair or directional scenario, the explanation output can report the final recommendation, primary reason, mechanism evidence, model evidence, site context needed, and site-validation requirements. This is intentionally more useful for engineering review than a black-box probability alone.

- Primary reason: a concise engineer-readable explanation such as model disagreement, protected bulk acid/base RTD pattern, non-thermal dominance, phase/fouling concern, fast pressure/kinetics concern, or no RTD-creditable thermal mechanism identified.
- Mechanism evidence: reaction type, severity, heat/gas/toxic/pressure/corrosion flags, v1.2b proxy flags, directionality sensitivity, and stored-service reliability review signals.
- Model evidence: model score, model disagreement with the calibrated weak-label layer, and near-boundary behavior.
- Site context needed: stored/incoming direction, tank inventory, concentration, order of addition, credible misdirected volume, RTD location, mixing quality, fouling/coating history, and pressure-relief context where applicable.

7.2 Service-Level Chemistry Basis for Retained RTD-Candidate Scenarios

The final notebook also includes an enhanced service-level explanation layer for retained RTD-candidate scenarios. This layer was added because a tank/service recommendation should not only say how many RTD-candidate scenarios exist; it should also explain which pair/direction scenarios created the candidate status and why those scenarios were retained from a chemistry and safeguard-creditability standpoint.

For each retained RTD-candidate scenario, the enhanced output reports:

- source severity basis;
- heat-generation basis;
- reaction type / mechanism class;
- positive RTD pathway;
- why RTD could detect the event;
- failure modes checked and not found;
- remaining site validation needed;
- and chemistry-basis strength.

The chemistry-basis strength field intentionally separates stronger retained cases from weaker screening-retained cases.

A strong positive chemistry basis means the retained scenario has a specific bulk-thermal chemistry basis, such as a bulk liquid acid/base exotherm, protected acid/base RTD pathway, or calibrated RTD-positive pattern.

A weaker screening-retained basis means the scenario was retained because the source compatibility screen showed high consequence plus heat generation and the available logic did not identify a dominant reason to eliminate RTD. These scenarios should not be treated as definitive RTD recommendations; they require stronger engineering review before final RTD credit.

This distinction is important because high consequence plus heat generation is only a screening trigger. It is not sufficient by itself to justify RTD. Final RTD credit still requires validation of actual chemical mapping, stored/incoming direction, tank inventory, concentration, credible misdirected volume, mixing, RTD location, response basis, and stored-service reliability.

8. Evaluation & Validation Strategy

Evaluation must match the revised capstone objective. The most important question is not whether the model can perfectly reproduce a six-class severity label. The most important question is whether the model produces useful, conservative, and explainable mitigation recommendations.

Metric	Definition
LF coverage	How often each LF produces a nonblank vote.
LF precision when voting	How often a rule agrees with the audited target when it actually votes. Abstentions are not counted as wrong.
LF conflict rate	How often active LFs disagree on the same row.
RTD recommendation precision	How often predicted RTD decisions agree with audited rtd_required decisions.
Review routing	Whether uncertain/conflicting cases are sent to peer review rather than forced into false certainty.
Stored-service reliability review	Whether tank recommendations respect stored-service RTD feasibility.
Severity-demotion flag quality	Reported as candidate-review performance only; not treated as a separate supervised model target in v1.
Prospective audit performance	Agreement of the frozen weak-label layer and model predictions against audit rows selected after the rules were frozen. Reported separately from development-set performance and later treated as calibration once v1.2b uses those audits.
Directionality review routing	Whether A_into_B / B_into_A or inventory-sensitive cases are correctly routed to tank-level review when site context is insufficient.
Class-imbalance metrics	Balanced accuracy, RTD precision, RTD recall, F1 score, average precision, and false negatives are emphasized over plain accuracy.
Grouped CV recall / false negatives	Primary safety-screening check: whether the model misses rare RTD-required weak-label cases under grouped cross-validation.
Average precision	Useful for rare positive classes because it compares ranked positive-class performance against baseline prevalence.
Model-challenge review count	How many rows the model would route to engineering review because it disagrees with the calibrated weak-label layer.
GitHub-safe reporting	Whether public artifacts communicate the workflow and results without exposing private audit workbooks or row-level audit notes.

8.1 Final Phase 1 Metrics

The final Phase 1 MVP should be interpreted as calibrated decision-support evidence rather than a statistically powered validation study. Key implementation metrics are summarized below.

Area	Metric	Final Phase 1 Value
Dataset	Public chemical-pair rows	313
Dataset	Directional scenario rows	626
Audit	Combined audited rows	36 total: 21 initial development + 15

		prospective stress-test
Audit	Audited RTD-required rows	5
Audit	Audited RTD-not-required rows	31
Training pool	Post-audit leakage-safe rows	271
Training pool	v1.2b RTD-required weak-label fraction	11 / 271 = 4.1%
v1.2b audit calibration	Precision / recall / F1	83.3% / 100.0% / 90.9%
v1.2b audit calibration	False positives / false negatives	1 / 0
Grouped CV model	Primary model	Evidence-weighted logistic regression
Grouped CV model	Balanced accuracy / precision / recall / F1	98.3% / 55.0% / 100.0% / 71.0%
Grouped CV model	Average precision / ROC-AUC	60.0% / 98.2%
Final pair overlay	Pair-level recommendations	279 RTD not required, 12 RTD required before directional validation, 22 model-challenge reviews

8.2 Validation Strength and Statistical Limitations

- The 36-row audited set supports MVP calibration, mechanism debugging, and prospective stress testing. It is not large enough to claim final statistical validation, especially because only 5 audited rows are RTD-required.
- The v1.2b model pool is highly imbalanced: only 11 of 271 leakage-safe training rows are RTD-required. For that reason, plain accuracy is not emphasized. Balanced accuracy, RTD precision, RTD recall, F1, average precision, ROC-AUC, and false-negative count are the relevant metrics.
- The primary model achieved 100% recall and zero false negatives in grouped cross-validation, but RTD precision was 55%. This is acceptable for Phase 1 screening because model-positive disagreements are routed to engineering review rather than automatically becoming installation scope.

9. Business Impact

The revised business case is mitigation optimization, not blanket severity reduction. This is more credible because the gold-set audit showed severity is often correct, while RTD applicability varies materially by mechanism and detectability.

- OpEx reduction: avoid unnecessary lab testing for high-confidence, well-understood cases.
- CapEx optimization: avoid blanket RTD installation where RTDs are not creditable or not the correct safeguard.
- Decision standardization: encode repeatable mechanism-aware engineering logic into a transparent framework.
- Portfolio prioritization: support future ranking of safeguards by risk reduction per dollar.
- Scenario realism: avoid one-size-fits-all recommendations by distinguishing pair-level chemical incompatibility from tank-level direction, inventory, and stored-service reliability review.

Screening-level savings estimates should be presented as directional. The model does not authorize safety scope reduction by itself; it provides auditable evidence for engineering review and detailed design.

9.1 Directional CapEx Avoidance Example

A representative screening scenario can be used to explain the potential business value without claiming finalized savings. In an initial blanket-scope concept, RTD installation on approximately 30-40 tanks was estimated at about $1.5MM in CapEx. The ChemRisk-AI Phase 1 workflow narrowed the RTD-candidate review set to roughly 10 services/tanks requiring engineering validation.

Using the initial estimate as a rough basis, this suggests potential avoided-scope opportunity on the order of $1.0MM-$1.1MM if engineering validation confirms that blanket installation is not required. This should be described as potential CapEx avoidance, not realized savings, until final project scope is approved.

The model does not authorize eliminating safeguards. It provides auditable evidence that helps engineers focus detailed review on the scenarios where RTD may be useful, creditable, and practical.

10. Tech Stack

Tool / Method	Role in Project
Python / Pandas / NumPy	Data processing, feature engineering, LF implementation, QA checks, and analysis.
Scikit-learn	Leakage-safe grouped cross-validation using logistic regression and random forest baselines; class weighting and evidence-tier sample weights.
Mechanism-aware labeling functions	Primary explainability layer: each RTD recommendation traces back to engineering rules, abstentions, conflicts, and calibrated failure-mode proxies.
Weak supervision / LF aggregation	Creates RTD-required, RTD-not-required, or Review labels from partial engineering rules while ignoring abstentions.
Matplotlib / GitHub figures	Portfolio-ready visuals for architecture, class imbalance, audit calibration, grouped CV metrics, pair outputs, directional outputs, and service-level recommendations.
OpenPyXL / Excel audit templates	Private development workflow for auditable gold-set workbooks, formula-driven audit fields, and prospective audit candidate review. Raw audit workbooks are not published.
Future explainability / demo	SHAP analysis and a Streamlit or CLI demo remain future enhancements; they are not required for the frozen Phase 1 MVP.

11. Roadmap and GitHub Readiness

1. Completed: finalized Phase 1 target as RTD usefulness / RTD recommendation rather than broad six-class severity prediction.

2. Completed: rebuilt Python helper features and RTD labeling functions with spreadsheet-to-Python parity checks.

3. Completed: implemented calibrated v1.1 and corrected v1.2b weak-label layers with explicit versioning.

4. Completed: selected and audited 15 prospective stress-test rows after v1.1 was frozen, then documented why the combined 36-row set is calibration data rather than independent validation.

5. Completed: trained leakage-safe grouped CV baseline models on v1.2b weak labels and selected an evidence-weighted logistic model for review routing.

6. Completed: expanded pair-level outputs into direction-aware A_into_B and B_into_A scenarios and aggregated stored-service recommendations.

7. Completed: exported model card, final summary tables, and eight GitHub presentation figures.

8. Current GitHub package: publish README, design doc, public notebook, figures, model card, summary outputs, and schema files only; omit private audit workbooks and raw row-level audit notes.

9. Future work: create a fully synthetic runnable demo dataset so external reviewers can execute the workflow without private files.

10. Future work: expand prospective audits beyond 36 rows, intentionally oversampling RTD-required and hard-review cases.

11. Future work: add SHAP or permutation-style explanations if deeper model-level feature attribution is useful, and build a Streamlit or CLI demonstration interface.

11.1 Future Product Roadmap

The GitHub package should be presented as the completed Phase 1 MVP, not as a production software product. The product roadmap explains how the project could become more broadly useful across sites and industries.

- Phase 1 - weak-supervised MVP: completed in this repository using structured chemical-pair data, corrected v1.2b labels, grouped CV, row-level explanations, and direction-aware service/tank aggregation.
- Phase 2 - synthetic demo app: create a fully synthetic dataset and a lightweight Streamlit or CLI interface that lets users select chemical classes and view recommendations plus explanations.
- Phase 3 - SDS ingestion and class mapping: add SDS/NLP or LLM-assisted extraction, CAS/synonym handling, pH and incompatibility extraction, and confidence-scored mapping from actual chemicals to generic mechanism classes.
- Phase 4 - multi-site validation: validate the workflow across multiple site portfolios, larger audited datasets, tank maps, inventory data, and lab or incident/outcome evidence where available.

12. Failure Modes & Limitations

- The combined 36-row audit set is useful for MVP calibration and active-learning stress testing, but it is not statistically powered validation; only 5 audited rows are RTD-required.
- Proxy features cannot replace measured kinetics, heat of reaction, viscosity, phase behavior, tank level, concentration, order-of-addition, or mixing data.
- Chemical-class rules may miss edge cases or overgeneralize within a class.
- Severity-demotion positives are sparse; severity-demotion outputs should be treated as review flags, not a mature predictive model.
- Stored-service reliability review is a lookup-based engineering proxy and requires site validation before use in real design.
- The pair-level model does not by itself resolve direction of reaction. Directionality is handled downstream through scenario expansion and review routing, and final design requires site context.
- The system must remain human-in-the-loop for safety-critical decisions.
- The public GitHub package is intentionally not fully runnable without private audit files; a synthetic runnable demo is future work.
- The grouped CV model predicts weak labels, not laboratory or incident outcomes. It should be used as a review overlay, not an autonomous safety decision-maker.
- Actual-chemical-to-generic-class mapping remains a product risk. In Phase 1 this mapping is handled through the structured dataset; a broader product would need controlled SDS extraction, confidence scoring, and expert review of uncertain mappings.

- The current workflow is useful as an auditable decision-support framework, but industry-wide deployment would require more chemical coverage, additional audited examples, and multi-site validation.

12.1 GitHub Publication Boundary

- The public repository should include the README, design document, public notebook copy, figures, model card, final summary tables, and schema documentation.
- The private source workbooks, gold-set audit workbooks, fully executed private notebook with raw audit outputs, and row-level audit notes should not be published.
- The repository is best described as a portfolio/reporting package. A fully synthetic runnable demo is an appropriate future enhancement, not a requirement for the Phase 1 MVP.

13. Conclusion

ChemRisk-AI is best framed as a mechanism-aware safeguard decision-support system. It does not claim to predict chemical truth from limited data or to solve reaction kinetics without site context. Instead, the final Phase 1 MVP uses weak supervision, audited calibration, leakage-safe tabular modeling, model-assisted review routing, and direction-aware tank/service aggregation to support a specific engineering decision: whether RTD installation is useful, creditable, and practical enough to carry forward for engineering validation.

Severity demotion remains valuable when it is credible, but the current evidence supports using it as a sparse candidate review flag rather than a full model target. The final project is strongest as an industrial AI portfolio piece because it demonstrates realistic problem framing, domain-specific weak supervision, audit feedback loops, leakage control, class-imbalance-aware evaluation, row-level explanation output, human-in-the-loop review routing, and explicit handling of directionality, inventory, stored-service limitations, and future productization needs.

Appendix A. Revision Log - What Changed and Why

This log is placed after the main document so readers first understand the problem, architecture, and decision logic before reviewing how the project evolved.

Revision	Reason
Moved revision log to appendix	Domain context should come before change history so readers are not confused by project-specific terms.
Added reader orientation	Defines C1-C6, C5/C6, RTD, weak supervision, labeling functions, gold set, and pair vs tank decisions for readers without chemical-process background.
Reframed primary target	Shifted from full severity prediction to pair-level RTD usefulness and tank-level RTD recommendation.
Clarified severity-demotion role	Severity demotion is now a secondary rule-based review flag, not a primary model target. SLF1 and SLF2 remain candidate flags; SLF3 is retired from v1.
Clarified dataset readiness	Current data supports a weak-supervised tabular MVP, not a first-principles chemistry predictor.
Added tank-level aggregation	Explains how pair-level predictions become storage-tank RTD recommendations using stored-service reliability review.
Updated architecture diagram	Diagram now shows severity as sparse review flags, RTD detectability as primary weak-supervision target, and tank-level aggregation as the final engineering decision layer.
Added calibrated v1.1 weak-label layer	Documents the separation between spreadsheet-parity LF reconstruction and targeted v1.1 calibration after development audit diagnostics.
Added prospective audit cycle	Captures that the next 15 audit rows are selected after v1.1 is frozen using model

	disagreement, uncertainty, and mechanism coverage gaps.
Added directionality and inventory-context layer	Addresses the engineering limitation that A_into_B can differ from B_into_A because stored inventory, dilution, concentration, and order-of-addition can affect mechanism and RTD detectability.
Updated modeling and evaluation plan	Adds leakage-safe grouped cross-validation, class-imbalance metrics, evidence-tier weighting, prospective audit performance, and directionality review routing.
Finalized corrected v1.2b layer	Documents the prospective-audit corrections for gas/pressure, fast-kinetics, phase/fouling, and protected bulk acid/base RTD cases.
Added final metrics and validation limitations	Clarifies that the 36-row audit set supports MVP calibration but not statistically powered production validation.
Clarified model-assisted review overlay	Explains that the tabular model routes disagreements to Review rather than automatically overriding mechanism-aware weak labels.
Updated stored-service terminology	Reframes stored-service practicality wording as RTD reliability review to avoid overstating surfactant, polymer, silicate, or emulsion services as automatically impractical.
Updated Tech Stack and future work	Moves SHAP and Streamlit to future enhancements and reflects what actually shipped in the Phase 1 GitHub package.
Added GitHub-safe publication guidance	Documents that private audit workbooks and raw row-level audit notes are omitted from the public repository while design docs, figures, model card, and summary outputs are included.
Added current MVP vs future product framing	Clarifies that Phase 1 is a decision-support MVP and that SDS ingestion, chemical class mapping, app deployment, and multi-site validation are future product steps.
Added LLM positioning	Explains why a general-purpose LLM may support SDS extraction but should not replace controlled, auditable decision routing for safety-critical safeguards.
Added row-level explanation layer	Documents the final notebook explanation output for primary reason, mechanism evidence, model evidence, and site-validation needs.
Added directional CapEx avoidance example	Frames the 30-40 tank / $1.5MM scenario as potential CapEx avoidance pending engineering validation, not realized savings.
Added product roadmap and mapping limitation	Documents the future synthetic demo, SDS ingestion, confidence-scored actual-chemical-to-generic-class mapping, and multi-site validation needs.
Added enhanced chemistry-basis explanation layer	Documents the service-level retained RTD-candidate explanation output, including source severity basis, heat-generation basis, positive RTD pathway, detectability basis, failure modes checked, remaining validation, and strong positive chemistry basis versus weaker screening-retained basis.